

Exercise 1:

1. If $q > 0$ then the magnetic force will be in the negative y-direction
2. If $q < 0$ then the magnetic force will be in the positive y-direction
3. If both fields are present, then sign of the charge doesn't matter for the total force can't be zero with these assumptions, as $\mathbf{v} \times \mathbf{B}$ and $\mathbf{E}$ both point in negative y-direction, and both are multiplied by the same charge. $\pm q \cdot (-E) + (\pm q) \cdot (-vB) \neq 0$

Exercise 2-6:

Exercise 2)

B) $A \cdot B = 1$ $AB \cos \theta = \sqrt{2} \cdot \sqrt{2} \cos \theta$
 $\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

D) $A \cdot B = 2$ $AB \cos \theta = \sqrt{2} \cdot \sqrt{3} \cos \theta$
 $\Rightarrow \cos \theta = \frac{2}{\sqrt{6}} \Rightarrow \theta \approx 35.26^\circ$

C) $A \cdot B = 1$ $AB \cos \theta = 3 \cos \theta$
 $\Rightarrow \cos \theta = \frac{1}{3} \Rightarrow \theta = 70.53^\circ$

Exercise 3: 1. a) $\nabla \phi(x, y) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}$

b) $\nabla \phi(1, 1) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ c) The gradient in b) shows $\frac{\partial \phi}{\partial x_1} \frac{\partial \phi}{\partial x_2}$ that ϕ changes by 2 in both x_1 and x_2 direction in the point (1, 1)

2. a) $\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} = 2$ in both x_1 and x_2 direction in the point (1, 1)

b) $\nabla \cdot \mathbf{F}$ is positive so it's a source. Means the vector field points out of itself like the charge of (b) x_3 direction and a positive particle. represents F's rotation.

Exercise 4)

a) $\nabla V_1(x_1, x_3) = \begin{pmatrix} 3x_1^2 \\ 2x_3 \end{pmatrix}$

b) $\nabla \cdot V_2 = 2x_1 + 2x_3$

c) $\nabla \times V_3 = \begin{pmatrix} x_1 \\ x_2 + x_3^2 \\ -x_1^2 \end{pmatrix}$ $\begin{pmatrix} \frac{\partial x_1}{\partial x_1} & \frac{\partial x_2}{\partial x_2} & \frac{\partial x_3}{\partial x_3} \end{pmatrix} \begin{pmatrix} 2 + x_2 x_1^2 \\ 0 \\ x_1 x_2 + x_1 x_3^2 \end{pmatrix}$

Exercise 5) $\int_0^1 F_1 dx_1 + \int_0^1 F_2 dx_2 + \int_0^1 F_3 dx_3$
 $\mathbf{F} = \begin{pmatrix} x_1^2 \\ -x_1 x_3^2 \\ -x_2^2 \end{pmatrix}$ $d\mathbf{l} = \begin{pmatrix} dx_1 \\ dx_2 \\ dx_3 \end{pmatrix}$
 $-\int_0^1 x_1^2 dx_1 + \int_0^1 x_1 x_3^2 dx_2 + \int_0^1 x_1^2 dx_1 + \int_0^1 -x_2^2 dx_3$
 $-\int_0^1 x_2^2 dx_3 - \int_0^1 x_1 x_3^2 dx_2$
 $\Rightarrow 0$

Exercise 6) Assuming cylindrical surface centred around the x_3 -axis - $\hat{n}_i = (0, 0, 1)^T$ and $\hat{n} = \left(\frac{x_1}{\sqrt{x_1^2 + x_2^2}}, \frac{x_2}{\sqrt{x_1^2 + x_2^2}}, 0 \right)$